LAB ASSIGNMENT REPORT

Resistors, LEDs, Potentiometer, LDR

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Task 1: LEDs Series Connection

In here first we took the red, green, yellow and blue led bulbs. Then in the breadboard we connected them in series. Connected the red led positive terminal series with yellow led negative terminal, then positive terminal of yellow led connected series with blue led negative terminal, then blue led positive terminal connected series with green led negative terminal. Then positive terminal of green led connected series with a resistor and it connected to positive side of battery. The negative terminal of red led is connected to negative side of battery. So with the power the four LEDs got brighten. If removed one led observed that all the other LEDs switched off.

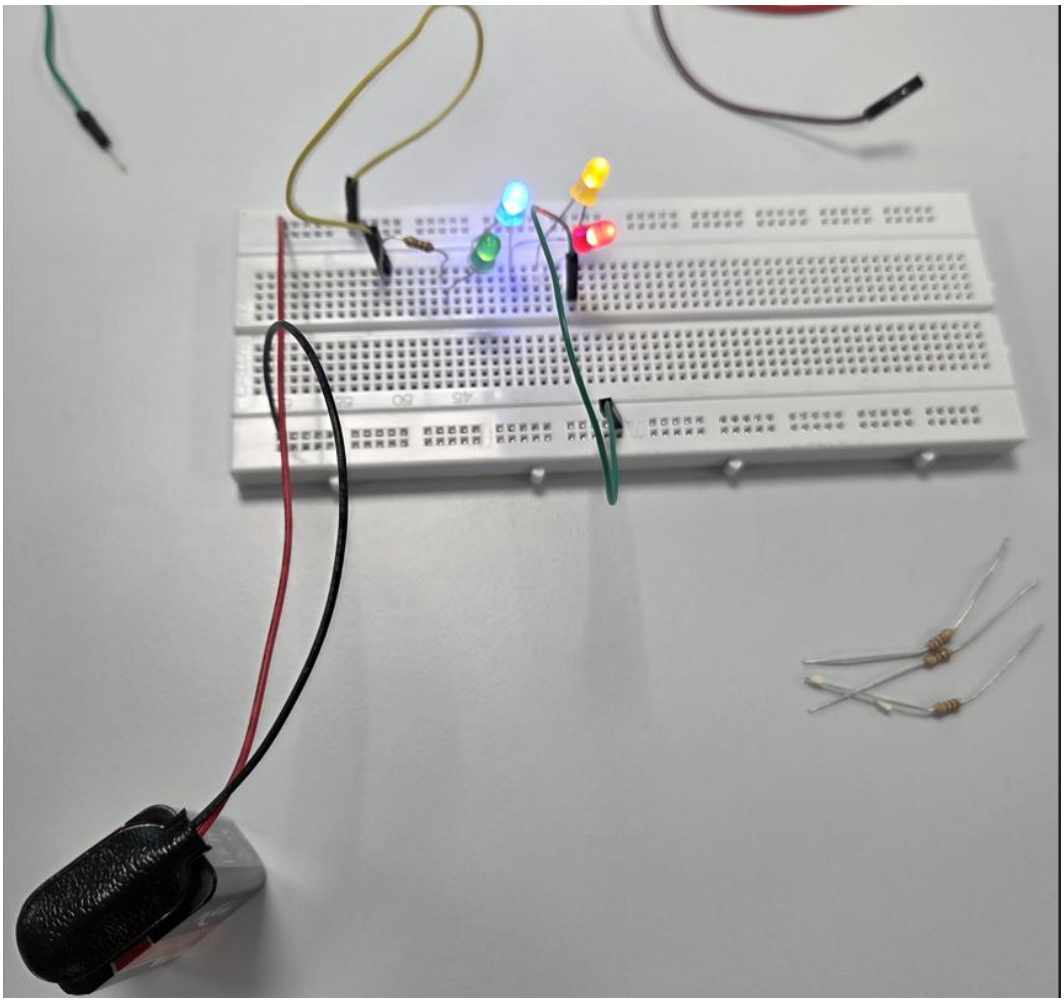

Figure 1: LEDs in Series

Task 2: LEDs Parallel Connection

In here the connection made with 4 LEDs, red, green, blue and yellow. Four resistors used. The connection made led bulbs separately with resistors. Positive side of LEDs connects to resistors and that

resistors connected with positive side of power source. Negative side of power source is connected to the negative terminal of the power source. Observation is all the four bulbs lighten up when gives the power. If removed one led from circuit other LEDs remains lighten.

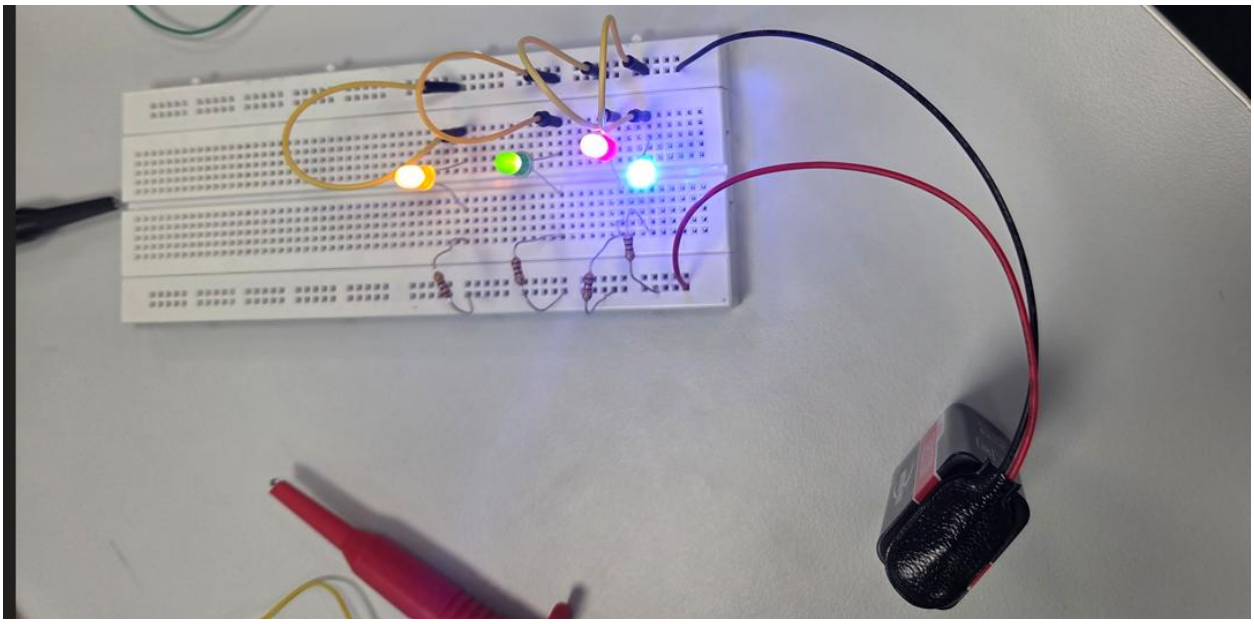

Figure 2: LEDs in Parallel

Task 3: Measuring the Resistance

In here connected the 4 resistors in series. The Ohm's law gives the resistance in a series circuit as R= R1+R2+….. . So the resistors connected series in the breadboard and the multi meter used to measure the resistance. Multi meter knob turns to Ohm symbol and the positive wire of voltmeter connects to one end of first resistor in series and the negative wire of multi meter connects the last resistor in series. Observation is the resistance shows in multi meter is similar to addition of resistance in each resistor.

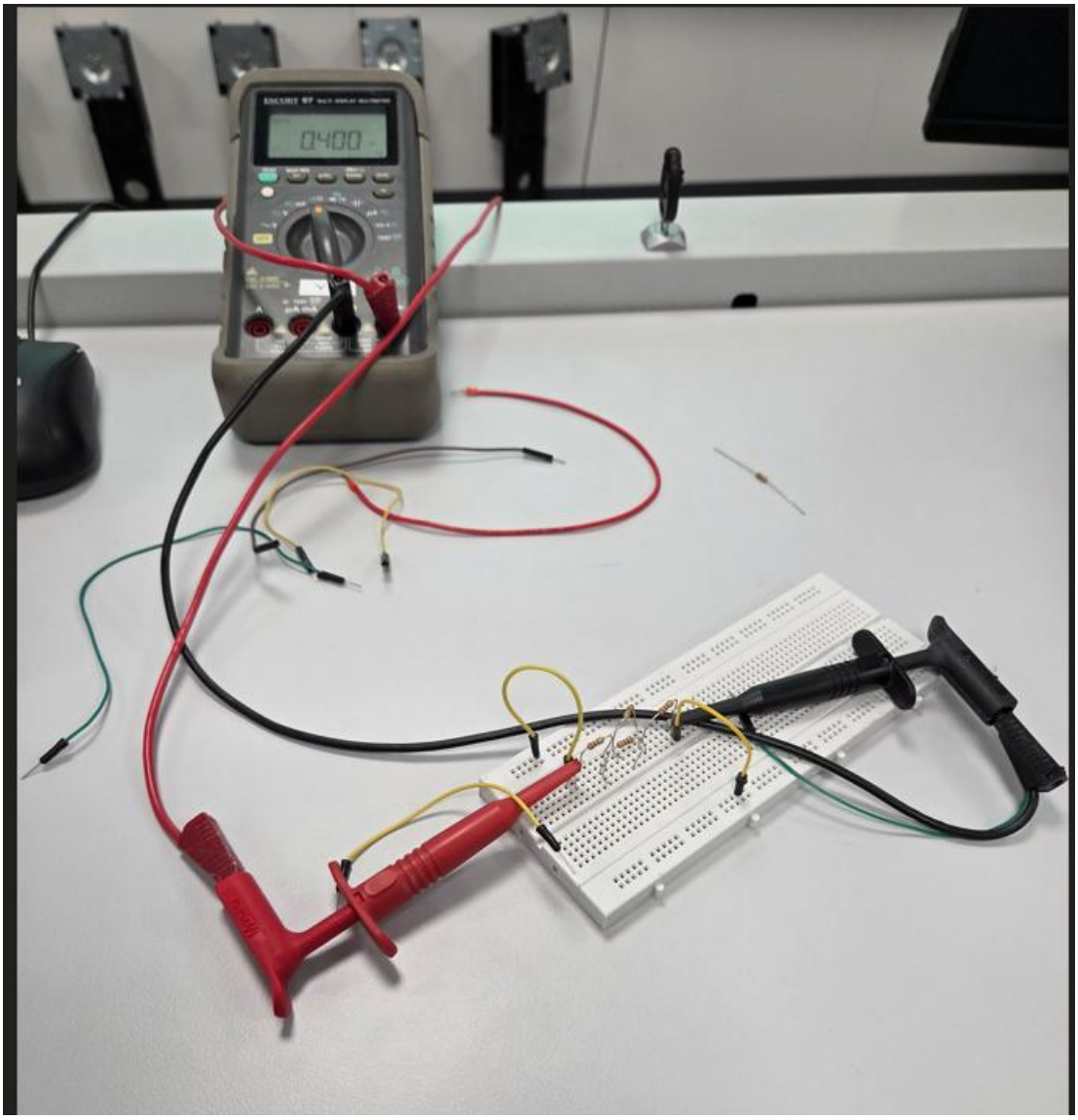

Figure 3: Measuring the Resistance

Task 4: Measuring the Voltage of LEDs

Two LEDs connected series in breadboard. The positive terminal of first LED in series is connected to a resistor and it connected to positive terminal of power source. The negative terminal of last LED in series is connected to negative terminal of power source. Then the multi meter used to check the voltage of LEDs. The knob in multi meter rotates to the voltage symbol and the positive wire of voltmeter is connected to positive side of LED and negative wire of voltmeter is connected to the negative side of LED. Then readings shoved in multi meter related to the LED voltage.

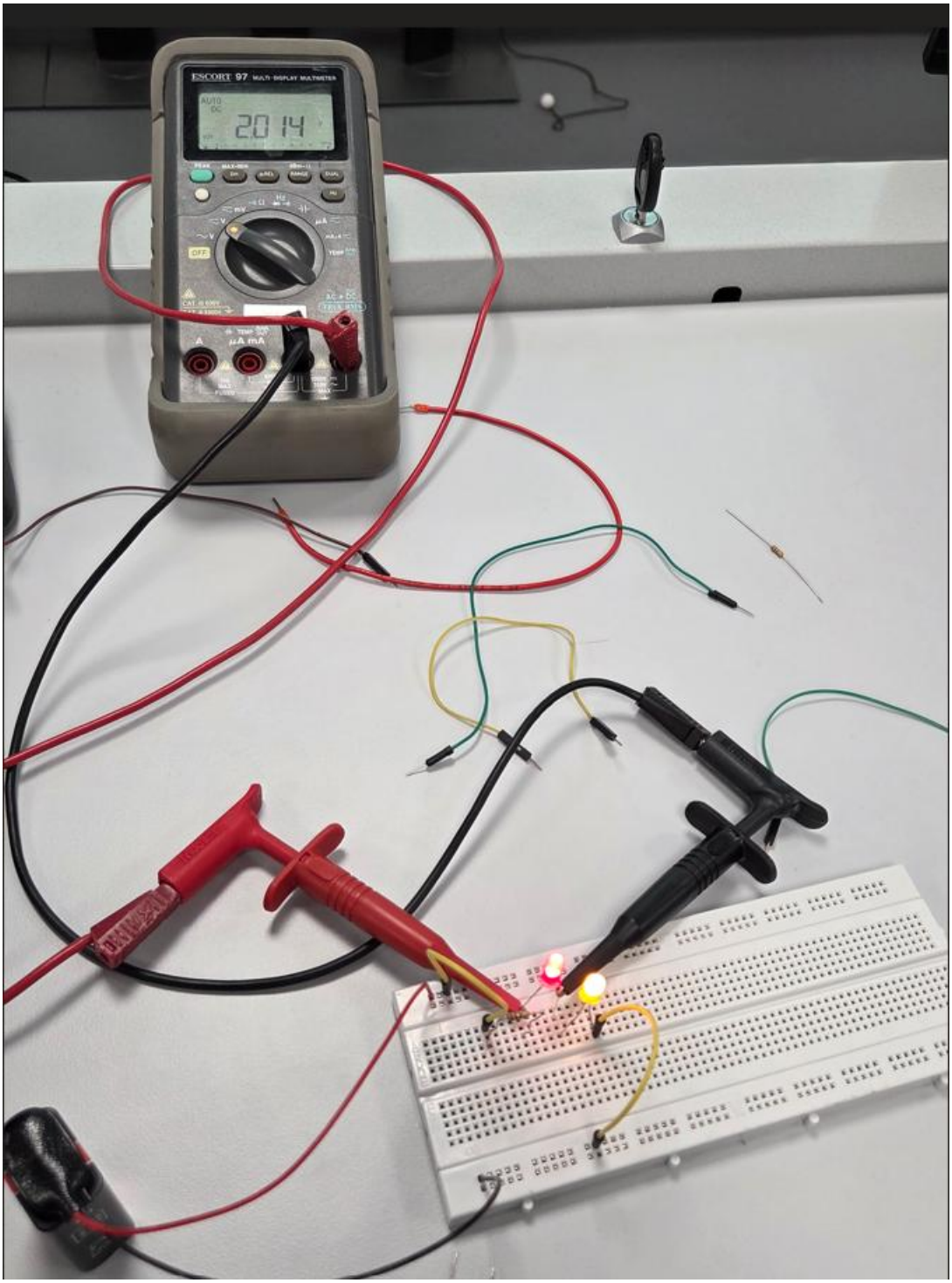

Figure 4: Red LED voltage

Task 5: Multiple LEDs With Push Buttons

In here used one push button, one LED. The push button positive terminal connected to the resistor and it connects to positive side of battery. The negative side connects to LEDs positive terminal and negative terminal of LED is connecting to negative side of battery. Observation is when press the push button the LED light up.

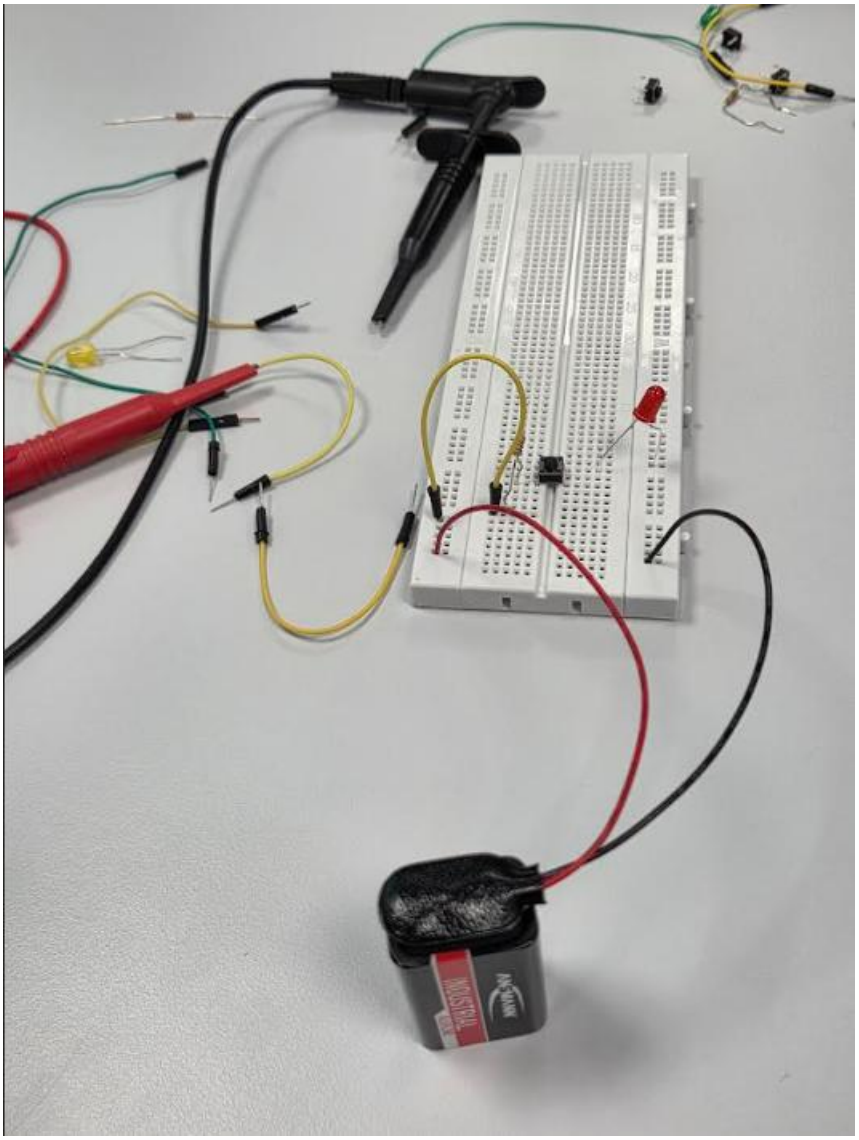

Figure 5: Before pressing push button

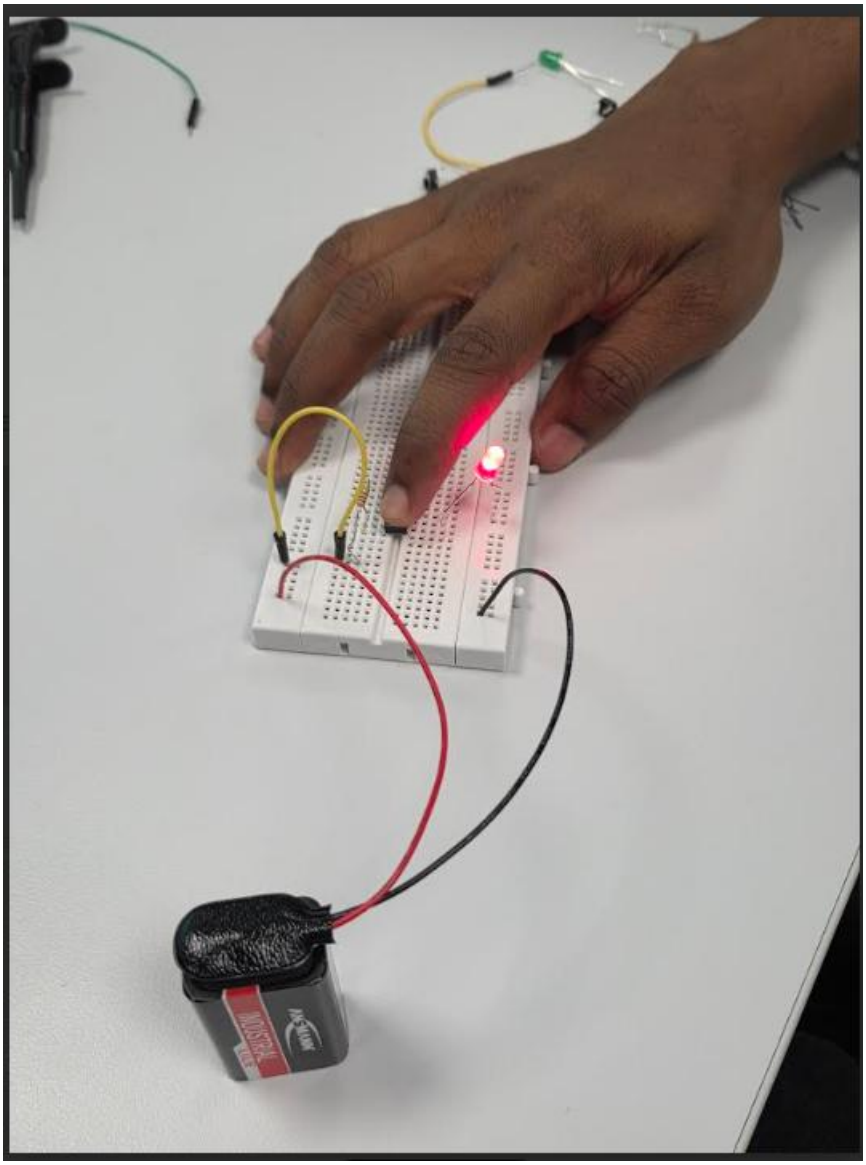

Figure 6: After pressing push button

Task 6: Turn Lights Off With Push Buttons

In here used four LEDs and four push buttons. Connected LEDs with push buttons but circuit short circuit by wires which connect with push buttons and negative terminal of the battery. Observation is when power up the circuit all the LEDs starts lighten up and when press the push button related to the LED, the related LED is switched off.

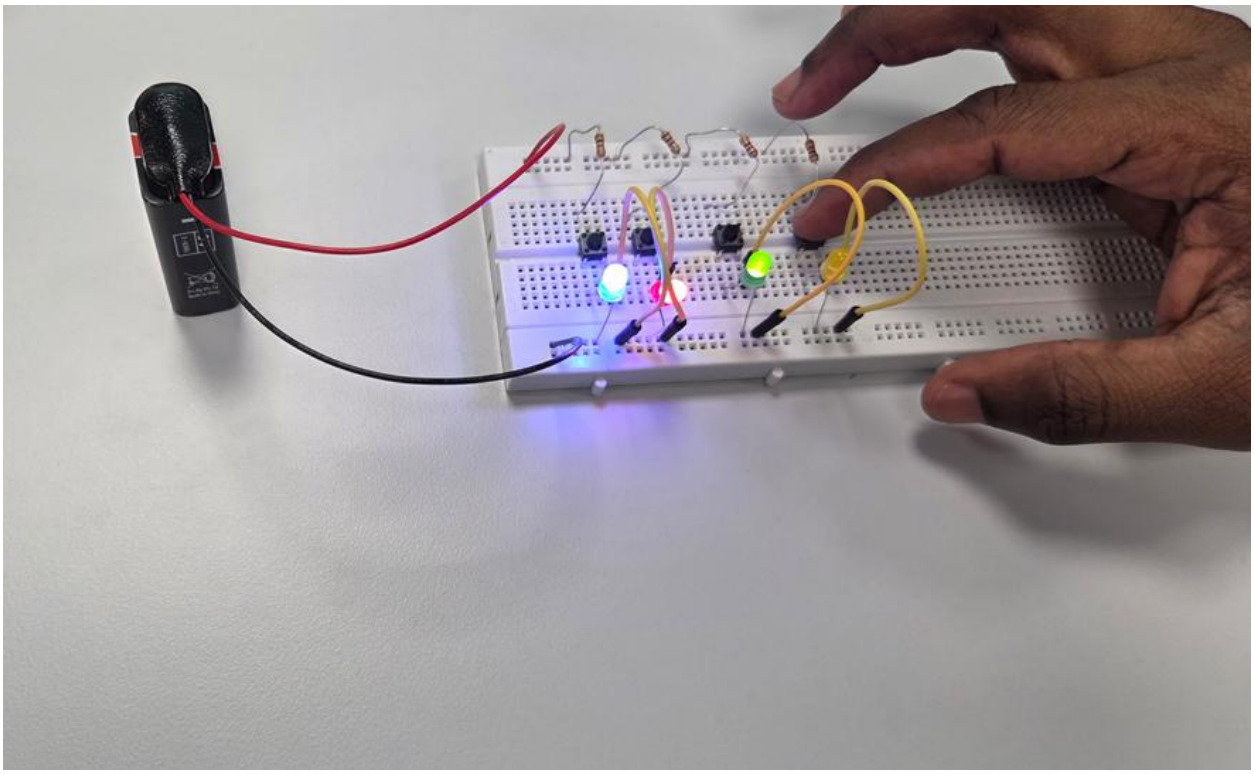

Figure 7: After pushing press button

Task 7: Potentiometer (Variable Resistor)

In here first identified the connection of the potentiometer as input, wiper and ground. Then connected the positive side of battery to input terminal, connected wiper and the ground. Took on output wire from wiper and connected it to a resistor. The other end of resistor connected to the positive terminal of LED and the negative terminal of LED is connected to the negative terminal of battery using wire. When power comes the lighten ability of LED increases and decreases as we rotate the knob of potentiometer.

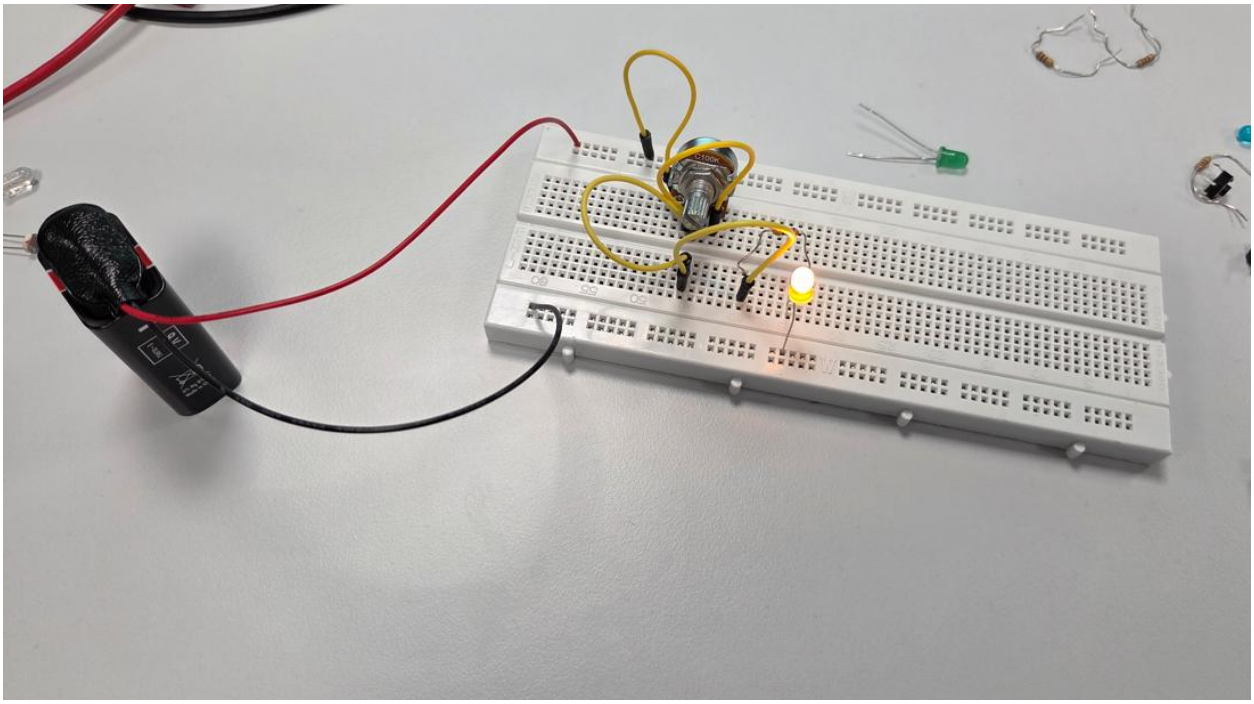

Figure 8: Potentiometer wiring

Task 8: Light Dependent Resistor (LDR)

The positive terminal of the battery is connected to the one end of resistor and the other end of the resistor is connected to the one terminal of LDR. The other terminal of LDR is connected with positive terminal of LED and the negative terminal of LED is connected to the negative side of the battery. Observation is when LDR is covered with hand the LED dim. So maximum resistance in dark and minimum resistance in light.

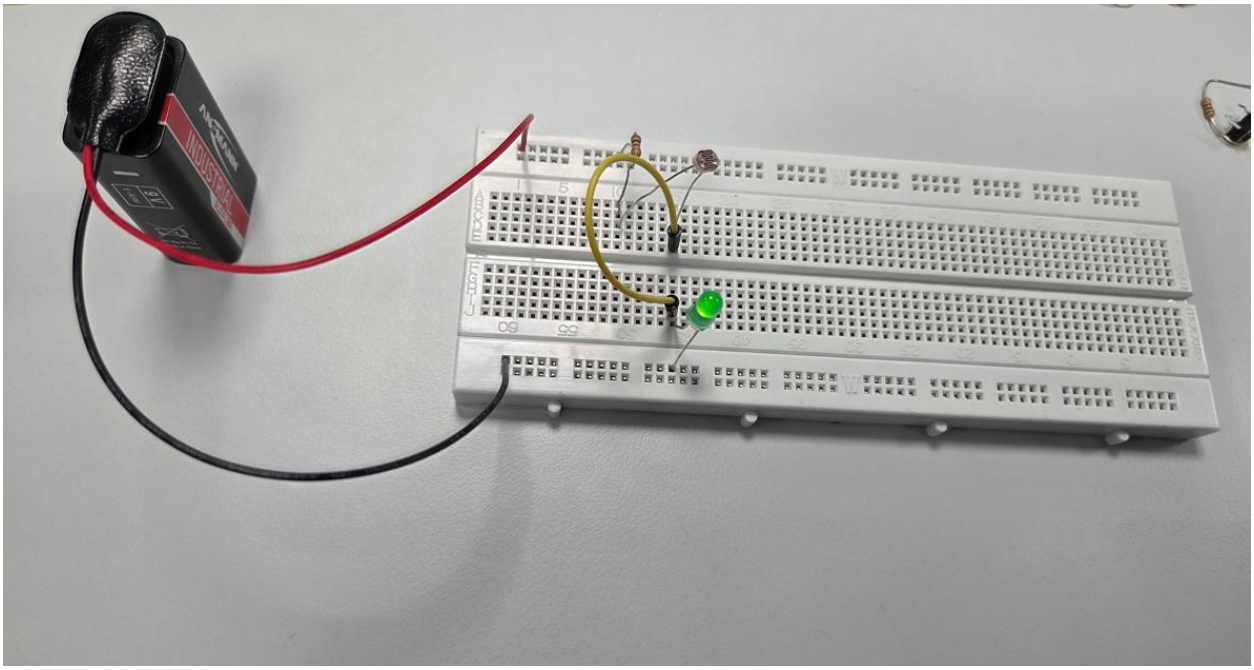

Figure 9: LDR in bright light

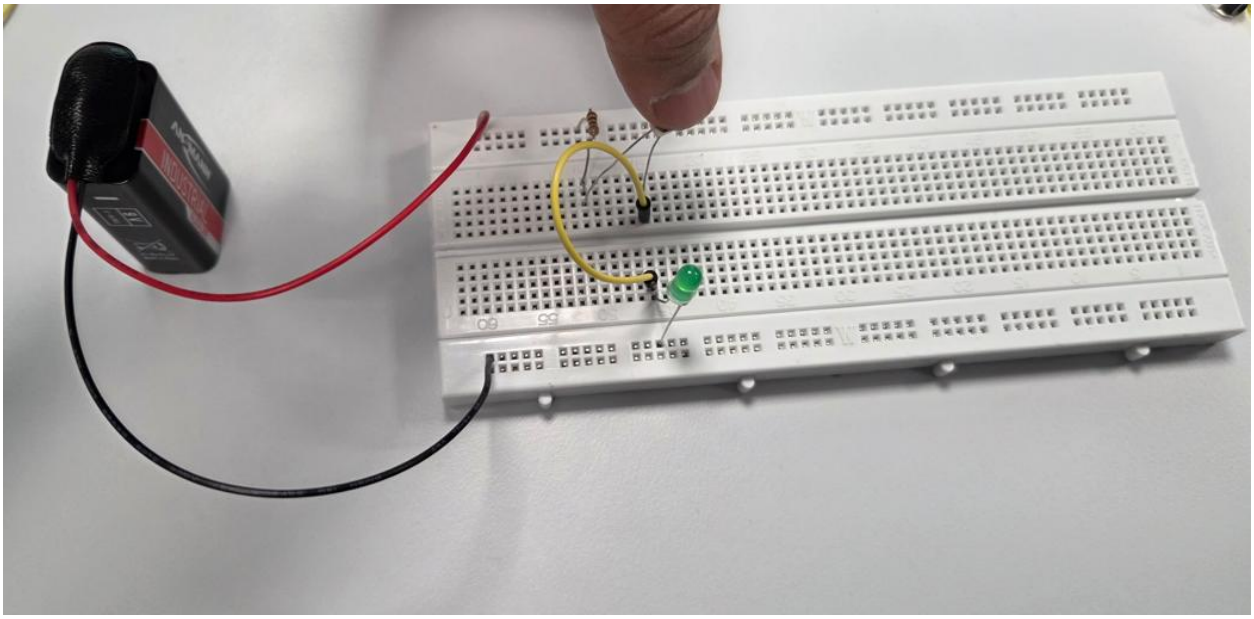

Figure 10: LDR in darkness

Task 9: Color the Rainbow

Used components as LDR, potentiometer, push buttons, resistors and tri color LED. The one push button connects with resistor, one push button connects with LDR and another resistor and one push button is

connected with potentiometer. The outputs of push buttons connected with the tri color LED and LED ground is connected to the negative side of the battery. Resistors and potentiometer connected to the positive side of the battery. Observations are when pressing push buttons the tri color LED lighten up, the controlling resistor of potentiometer and the LDR the color of light is changed.

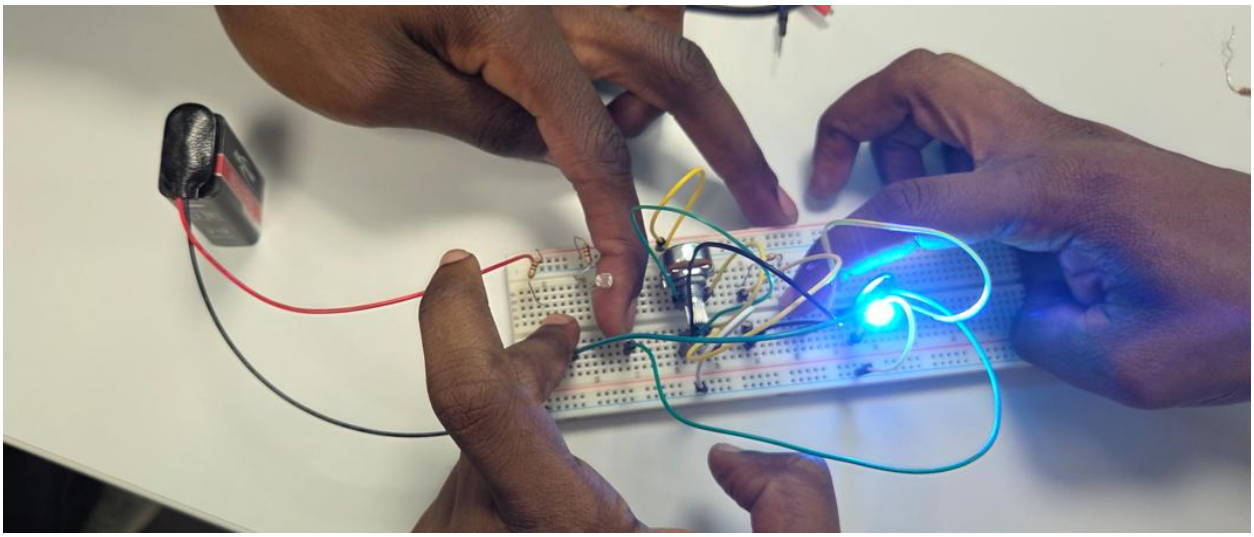

Figure 11: Pressing Push Buttons

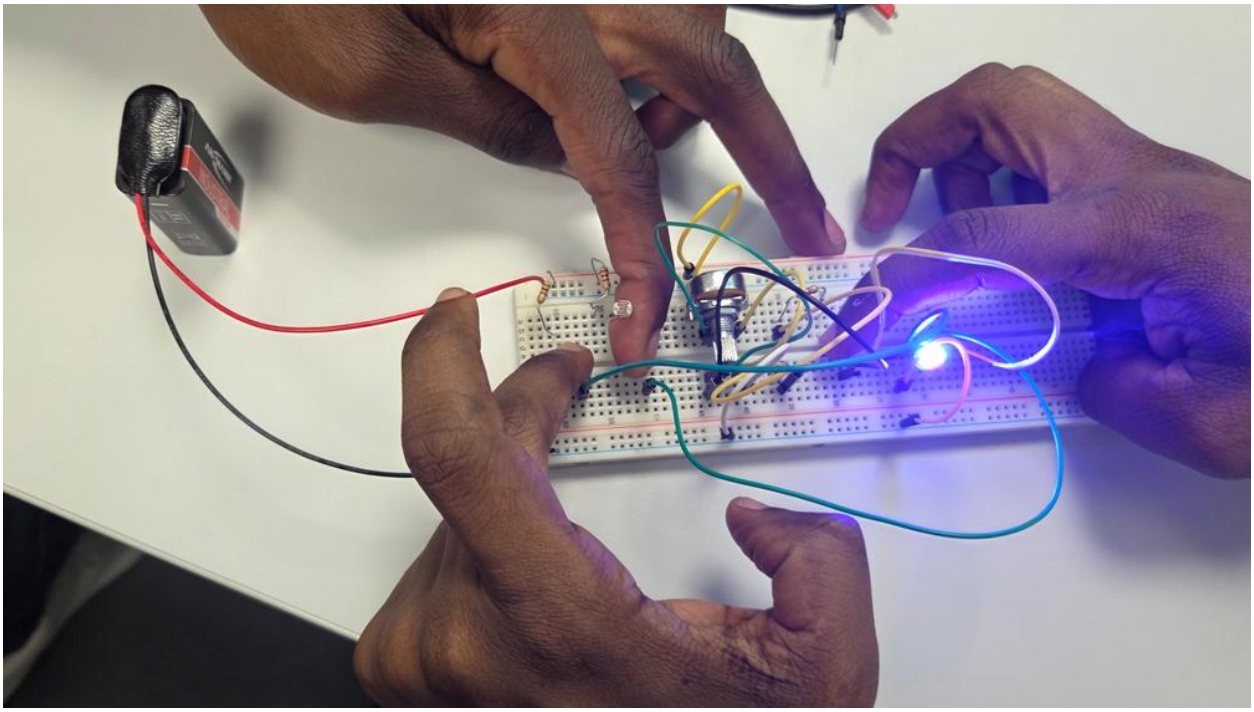

Figure 12: Controlling resistor and change the colors in LED